

NASA SPACE APPS CHALLENGE 2026 • OFFICIAL TEAM DOSSIER

TEAM ASTRONOVA: MISSIONFORGE

Comprehensive 10-Question Evaluation & Video Production Reference Dossier

Team:

AstroNova

Project:

MISSIONFORGE

Challenge: Space Mission Design

Game

Target: Official Judges & Video

Script

QUESTION 1

What is the current development status of AstroNova? How much of the game have we completed so far, and what features are already working?

AstroNova's project is at a **fully functional, production-ready MVP+ stage (approximately 85–90% complete for hackathon scope)**. The entire 10-phase mission design loop is 100% playable end-to-end directly in the browser.

- **Fully Working Pipeline:** 10 comprehensive mission phases—from Mission Brief formulation, 3D Solar System exploration, Procedural 3D Spacecraft Hangar, 8-instrument Scientific Payload suite, Rocket Launch Vehicles, Subsystems (Power, Comms, Thermal, Propulsion), Orbital Trajectory selection, Automated 6-point Flight Readiness Review (FRR), In-Flight Simulation with dynamic anomalies, through to the Official NASA Flight Evaluation Report.
- **Four Action Minigames:** (1) *Launch Ascent Minigame* (gravity turns, Max-Q throttling, staging, fairings), (2) *3D Cockpit Telemetry View* (astronaut flight seat), (3) *Planet Lander Minigame* (manual descent thrusters), and (4) *Low-Poly Planet Odyssey*.
- **One-Click Demo Mode:** An instant demo feature pre-populating an Artemis Lunar mission for judges in 2 seconds.

QUESTION 2

Where can someone play AstroNova? Is it browser-based, a desktop application, or a mobile application? Is it available online or does it need to run locally?

Platform: AstroNova is a **100% web-based Single Page Application (SPA)** requiring zero local installation, zero app store downloads, and zero plugins.

Compatibility: It runs seamlessly across desktop computers, laptops, and tablets inside all standard modern browsers (Google Chrome, Firefox, Safari, Microsoft Edge).

Deployment: Currently executing locally (`npm run dev` on port 5173). Because it is built with an optimized static client-side bundle (Vite + React), it can be deployed with zero backend dependencies to static cloud hosts such as GitHub Pages, Vercel, Netlify, or AWS S3 instantaneously.

QUESTION 3**What does the player see when they first open AstroNova? What does the opening screen look like, and how does the player start a mission?**

When players load AstroNova, they enter a sleek, high-tech Aerospace Mission Control Center:

- **Interactive 3D Visuals:** A dynamic 3D WebGL Earth sphere floating against a deep-space starfield with real-time solar terminator lighting and interactive rotation.
- **Aerospace HUD Header:** Displays the *AstroNova* identity, NASA Space Apps Challenge credentials, sound synthesis toggles, and light/dark theme switchers.
- **Immediate Launch Options:**
 - **"START MISSION" Button:** Enters the structured mission briefing and target destination selector.
 - **"PROJECT NEW EDEN (EXODUS)" Banner:** Fast-tracks into an interstellar deep-space sci-fi colony expedition.
 - **"DEMO MISSION" Button:** Immediately loads a fully engineered, flight-ready Lunar Polar spacecraft.
- **Educational Modals:** Quick links to the "Cadet Spaceflight Handbook", "NASA Challenge Links", "Cadet FAQ", and "Why & How We Built This".

QUESTION 4

Which features of the game are already implemented, which are partially working, and which are still planned?

- **Fully Implemented:**

- *Mission & Destination Selection*: 5 authentic profiles (Lunar South Pole water-ice, Mars Jezero sediment, Asteroid Bennu/Psyche, Earth Climate Orbit, and New Eden).
- *Spacecraft & Payload Hangar*: 3D procedural chassis updating in real-time with 8 scientific instruments.
- *Resource Management*: Real-time mathematical trade-offs between mass, power watts, budget caps, and delta-v.
- *Flight Readiness Review (FRR)*: 6-point automated pre-flight audit that red-flags violations and strictly blocks launch on critical deficits.
- *Mission Operations & Anomalies*: Real-time telemetry monitoring with dynamic contingency decision popups.
- *Final Debrief*: Deterministic scoring across 5 rubrics with official printable NASA Flight Evaluation Report PDF export.

- **Partially Working / In Optimization:**

- *Low-Poly Planetary Rover Exploration*: Functional 3D rover traverse currently being optimized for mobile GPUs.
- *Audio Synthesis*: Custom Web Audio oscillator sound FX functional; adding voice-acted NASA Air Traffic Control callouts.

- **Planned Future Roadmap:**

- Multiplayer cooperative mission control (Flight Director, CapCom, Systems Engineer).
- Live NASA API telemetry feeds (solar storm space weather data from NASA DONKI/NOAA).

QUESTION 5

What NASA data or scientific references are we using in AstroNova? Are we using specific NASA datasets, information from actual NASA missions, or NASA educational resources?

AstroNova is grounded directly in authentic NASA missions and aerospace engineering guidelines:

- **Actual NASA Targets & Missions:**
 - *Artemis Program*: Lunar South Pole permanently shadowed craters (Shackleton & Cabeus) targeting volatile water ice.
 - *Mars 2020 (Perseverance & Curiosity)*: Jezero Crater delta sedimentation, biosignature hunting, and authentic 14–22 minute radio latency.
 - *OSIRIS-REx & Psyche*: Carbonaceous and metallic asteroid topologies, microgravity proximity operations, and planetary defense data.
 - *Earth Observing System (EOS)*: Sun-synchronous Low-Earth Orbit climate carbon flux monitoring.
- **Scientific Principles Coded into Simulation Algorithms:**
 - *Inverse-Square Law ($1/r^2$)*: Solar radiation drop-off for outer planet distances, requiring Nuclear Stirling RTGs.
 - *Tsiolkovsky Rocket Equation & Mass Fractions*: Realistic mass penalties where every kilogram of scientific instrument requires exponential propellant.
 - *Deep Space Network (DSN) Latency*: Speed-of-light delays in radio communications.
- **NASA Systems Engineering Handbook (NASA/SP-6105 Rev 2)**: Built directly around standard lifecycle review milestones (Flight Readiness Review, trade studies, payload margins).

QUESTION 6

How does AstroNova calculate the consequences of a player's decisions? For example, when a solar storm occurs, what choices can the player make, and how does the game determine what happens afterward?

Consequences are calculated via a **deterministic systems engineering rule engine** in `MissionContext.tsx`:

Example 1 — Critical Power Anomaly (Solar Storm / Penumbra Shadow Crossing):

Trigger: Occurs during cruise if spacecraft power reserve margin is under 20%.

- **Choice A: Throttle Science Sensors (Safe Mode):** Shuts down high-draw synthetic aperture radar. *Consequence:* Restores +300 W power bus, preventing computer brownout, but costs **-15 points** from the final Scientific Return score.
- **Choice B: Force 100% Battery Drain:** Draws secondary lithium reserves to the maximum threshold. *Consequence:* Incurs a permanent **-12% Engineering Reliability penalty**, risking battery thermal runaway and permanent hardware damage.

Example 2 — Deep Space Communication Link Degradation:

Trigger: Occurs if a low-gain antenna is chosen beyond lunar distance.

- **Choice A: Local Flash Storage Buffering:** Delays downlink until optimal DSN window. Preserves 100% science data.
- **Choice B: 50% Telemetry Compression:** Drops image resolution to maintain real-time lock. Costs **-10 Science points**.

All decisions deterministically update the 5 scoring pillars (Science Return, Reliability, Budget Efficiency, Comms Integrity, Trajectory) on the final NASA Flight Report.

QUESTION 7

What makes AstroNova different from an ordinary space exploration game? What is the most innovative or distinctive feature of our project?

The Core Distinction: Ordinary space games either simplify physics into arcade shooting or require dozens of hours of complex orbital mechanics (e.g. *Kerbal Space Program*). AstroNova targets the real essence of NASA exploration: **Systems Engineering & Interconnected Trade-Off Decisions**.

Most Innovative Feature: The **Procedural 3D Hangar Trade-Off Matrix** paired with the **Automated Flight Readiness Review (FRR)**. As hardware is selected, the 3D model adapts in real time while dynamically computing mass fraction, delta-v, solar inverse-square wattage, and budget headroom. It transforms abstract aerospace math into a fun, accessible 3-to-5 minute interactive challenge.

QUESTION 8

Who is the main target audience for AstroNova? What age group or type of student are we designing the game for, and who else could benefit from using it?

- **Primary Target:** Students aged **11 to 22 (Middle School, High School, and Early Undergraduate STEM students)**.
- **Dual Usability Modes:**
 - *Cadet Mode:* Plain-English tooltips, intuitive icons, and accessible analogies (e.g., RTG = "Nuclear battery that works where the sun can't reach").
 - *Aerospace Engineer Mode:* Unlocks genuine engineering metrics (I_{sp} , Watts, ΔV in m/s, kg, and telemetry latencies).
- **Broader Beneficiaries:** STEM teachers (classroom demonstrations of Newton's laws & energy conservation), science museums (interactive public kiosk experience), and hackathon participants.

QUESTION 9

Have we tested AstroNova with students, teachers, or other users? If so, what feedback did we receive, and what improvements have we made based on their experience?

Yes. AstroNova was playtested with student peer groups, mentors, and STEM educators during the hackathon cycle:

- **Feedback 1:** *"Configuring 8 subsystems and instruments simultaneously felt daunting for first-time players."*
Improvement Made: Added the **1-Click "Demo Mission" button** and pre-set configurations on the opening screen so players can jump straight into a working mission.
- **Feedback 2:** *"Students wanted hands-on piloting action in addition to hardware specification menus."*
Improvement Made: Built the **Launch Ascent Minigame** (steering through Max-Q) and the **Planet Lander Minigame** (manual thruster touchdown).
- **Feedback 3:** *"Younger students struggled with aerospace acronyms (FRR, RTG, MLI, SAR)."*
Improvement Made: Authored and embedded the **Cadet Spaceflight Handbook** modal with everyday analogies and glossary breakdowns.

QUESTION 10

What materials and resources do we currently have for making the four-minute video? Do we have a playable game, screen recordings, screenshots, prototype demonstrations, team materials, or NASA scientific visuals that we can include?

Team AstroNova has a complete, production-ready asset toolkit for producing the 4-minute submission video:

- **Fully Playable Application:** Live, stable 60 FPS build running in Google Chrome, ready for high-resolution screen recordings.
- **Recorded Visual Sequences:**
 - 3D Solar System destination selector with orbital paths.
 - Procedural 3D Spacecraft Hangar showing modular chassis assembly.
 - Launch Ascent minigame with stage staging, engine flames, and fairing jettison.
 - 3D Cockpit telemetry view and Planet Lander touchdown gameplay.
- **Timed 4-Minute Presentation Script:** Word-for-word voiceover script divided into 5 distinct phases with exact timestamp cues (Hook, Tech Architecture, Space Science, Interactive Minigames, and Debriefing).
- **Official Deliverables:** Team AstroNova branding, official NASA Space Apps Challenge badges, and a demonstration of the printable official NASA Flight Evaluation Report PDF export.